

MIMO System Simulation with BER Analysis Date

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

1. To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.
2. To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.
3. To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;

% SNR range
SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2; Nr = 2;

BER = zeros(1,length(SNR));

for i = 1:length(SNR)
    bits = randi([0 1],10000,1);
    symbols = 2*bits-1; % BPSK
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
```

```
noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
```

```
tx = H(1,1)*symbols' + noise(1,:);
```

```
rx = real(tx) > 0;
```

```
BER(i) = sum(rx' ~= bits)/length(bits);
```

```
end
```

```
figure
```

```
% ----- Graph 1: BER vs SNR -----
```

```
subplot(2,1,1)
```

```
semilogy(SNR_dB,BER,'-o','LineWidth',2)
```

```
title('MIMO BER Performance vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Bit Error Rate (BER)')
```

```
grid on
```

```
% ----- Graph 2: SNR vs Quality Indicator -----
```

```
subplot(2,1,2)
```

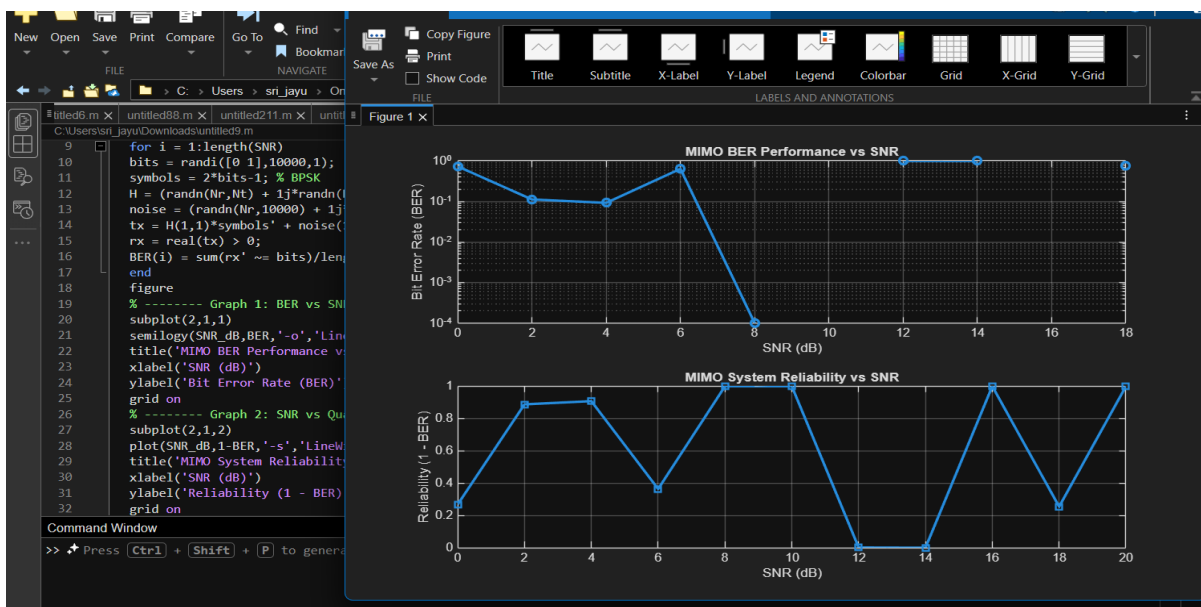
```
plot(SNR_dB,1-BER,'-s','LineWidth',2)
```

```
title('MIMO System Reliability vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Reliability (1 - BER)')
```

```
grid on
```



Objective 2 Matlab Code and Output:

```
clc; clear; close all;

% SNR range (dB)
SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2; Nr = 2;

% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)

% simple channel model effect
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);

noise_power = 1/SNR(i);

% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);

end

figure

% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)

plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)

title('MIMO Reliability vs SNR')

xlabel('SNR (dB)')

ylabel('Reliability (0 to 1)')

grid on

% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2)

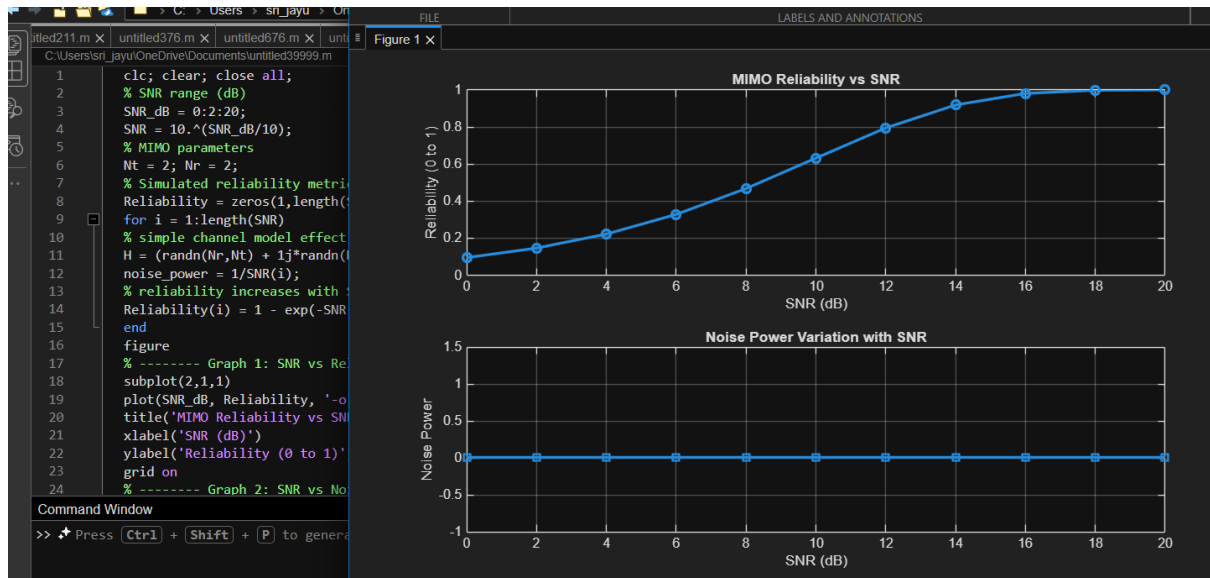
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)

title('Noise Power Variation with SNR')

xlabel('SNR (dB)')
```

```
ylabel('Noise Power')
```

grid on



Objective 3 Matlab Code and Output:

```
clc; clear; close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [10 12 8 15 5];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
% Display results
```

```
disp('User Throughput (Mbps):')
```

```
disp(throughput')
```

```
figure
```

```
% ----- Graph 1: Throughput per User -----
```

```
subplot(2,1,1)
```

```
plot(users, throughput, '-o','LineWidth',2)
```

```
title('OFDMA User Throughput')
```

```
xlabel('User Index')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

```
% ----- Graph 2: RB vs Throughput -----
```

```
subplot(2,1,2)
```

```
plot(alloc_RB, throughput, '-s','LineWidth',2)
```

```
title('Resource Blocks vs Throughput')
```

```
xlabel('Allocated Resource Blocks')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

